CHAPTER 25 · DERIVATIONS

QUANTUM PHYSICS

6 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 DE BROGLIE WAVELENGTH IN TERMS OF KINETIC ENERGY

SYLLABUS 25.2

RESULT $\lambda = \dfrac{h}{\sqrt{2mK}}$

1. de Broglie relation: $\lambda = \dfrac{h}{p}$, with momentum $p = mv$.
2. Square the momentum: $p^2 = m^2v^2 = m(mv^2)$.
3. Kinetic energy $K = \frac{1}{2}mv^2$, so $mv^2 = 2K$.
4. Substitute mv^2: $p^2 = 2mK$, so $p = \sqrt{2mK}$ and $\boldsymbol{\lambda = \dfrac{h}{\sqrt{2mK}}}$.

2 DE BROGLIE WAVELENGTH OF A PARTICLE ACCELERATED THROUGH A POTENTIAL DIFFERENCE

SYLLABUS 25.2

RESULT $\lambda = \dfrac{h}{\sqrt{2meV}}$

1. de Broglie relation: $\lambda = \dfrac{h}{\sqrt{2mK}}$.
2. A particle of charge e accelerated from rest through a potential difference V: kinetic energy gained = loss of electric potential energy, $K = eV$.
3. $\boldsymbol{\lambda = \dfrac{h}{\sqrt{2meV}}}$.

3 RADII OF BOHR ORBITS

SYLLABUS 25.3(B)

RESULT $r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}$

1. The electric attraction provides the centripetal force: $\dfrac{e^2}{4\pi\varepsilon_0 r^2} = \dfrac{mv^2}{r}$ … (1)
2. Bohr's postulate (quantised angular momentum): $mvr = \dfrac{nh}{2\pi}$, so $v = \dfrac{nh}{2\pi mr}$ … (2)
3. Substitute (2) into (1): $\dfrac{e^2}{4\pi\varepsilon_0 r} = mv^2 = \dfrac{n^2h^2}{4\pi^2mr^2}$.
4. $\boldsymbol{r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}}$; $r_1 = 5.3 \times 10^{-11}$ m.

4 ENERGY LEVELS OF HYDROGEN

SYLLABUS 25.3(C)

RESULT $E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}$

1. From (1): $KE = \frac{1}{2}mv^2 = \dfrac{e^2}{8\pi\varepsilon_0 r}$.
2. $PE = -\dfrac{e^2}{4\pi\varepsilon_0 r}$.
3. Total $E = KE + PE = -\dfrac{e^2}{8\pi\varepsilon_0 r}$.
4. Substitute r_n: $\mathbf{E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}} = -\dfrac{13.6}{n^2}$ eV.

5 MINIMUM X-RAY WAVELENGTH

SYLLABUS 25.4(C)

RESULT $\lambda_{min} = \dfrac{hc}{eV}$

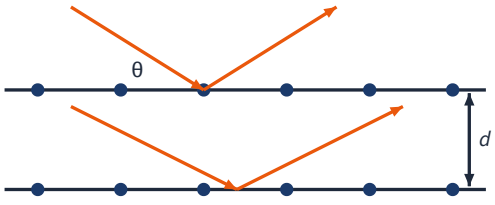
1. Electrons accelerated through p.d. V gain kinetic energy eV.
2. The shortest wavelength is emitted when all of this energy goes into one photon: $eV = hf_{max} = \dfrac{hc}{\lambda_{min}}$.
3. $\mathbf{\lambda_{min} = \dfrac{hc}{eV}}$ (independent of the target metal).

6 BRAGG'S LAW

SYLLABUS 25.4(E)

RESULT $2d \sin\theta = m\lambda$

1. X-rays are reflected by two adjacent atomic planes a distance d apart, at glancing angle θ.
2. The ray from the lower plane travels an extra path $2d \sin\theta$.
3. Constructive interference when the extra path is a whole number of wavelengths: $\mathbf{2d \sin\theta = m\lambda}$.



extra path of lower ray = $2d \sin\theta$